

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

JavaScript, TypeScript, Python, C++, Java, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services & queues, WebSocket / real-time streaming, hardware integration

Frontend & Desktop

React, Electron, cross-platform desktop UI, audio/video stream handling, HTML/CSS

Testing & DevOps

pytest, integration & regression testing, QA, LambdaTest, Git, GitHub Actions, CI/CD, Docker, Linux

Cloud & Data

AWS, GCP, Azure, MongoDB, PostgreSQL, Redis, SQLite

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a **full-stack internal tool (React, Node.js, MongoDB)** used daily by the AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved throughput by **60%** through cross-functional collaboration.
- Engineered a **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions**, integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Owned backend reliability and **QA** across the eTriage stack: monitored production logs during feature releases, shipped fixes for tenant-specific incidents, added integration tests, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect sensitive data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

Sonare - Cross-Platform Immersive Media Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform desktop application** integrating camera, microphone, frontend UI, and backend services into an on-device **real-time media pipeline** (MediaPipe hand tracking, whisper.cpp speech recognition, local Text-To-Speech), prototyped, designed, and shipped end-to-end under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** with async APIs and queues to keep the UI responsive while handling live **video and audio streams** locally on device, fully offline with no external network dependencies.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, Brave Search, SQLite

- Built a multi-stage **FastAPI** backend service that turns open-ended queries into structured tables, with deterministic parsers before LLM fallback and cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback for reliability, pipeline instrumentation, and a reviewer-facing trust **UI** for inspecting output quality.

cf-ai-research-scout - Real-Time Streaming Research Assistant

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers, WebSockets, SQL

- Built a research assistant that fetches web sources, generates multi-step digests, and **streams responses to the client over WebSocket in real time**, prototyped end-to-end on Cloudflare Workers with embedded SQL storage.
- Designed durable workflow boundaries so multi-step tasks survive process restarts and individual tool failures; state and conversation history persist server-side without an external database.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.